DrainQA

Social Science Notes — Chapter 15: Drainage System (Exercise Answers)

tags: #Class9 #SocialScience #Chapter15 #ExerciseAnswers #ObsidianNotes #GSEB

> Answer Categorization Standard
>
> - **Long Answers (Q1.6, Q2.1, Q2.3):** 8 points each
> - **Brief Answers (Q1.1, Q1.2, Q1.3, Q1.4, Q2.2):** 6 points each
> - **Short Answers (Q1.5):** 4 points each
> - **MCQs (Section 3):** Direct options with correct answers

---

1. Write short answers for the following questions

(1) Give difference : Himalayan rivers - Peninsular rivers

Answer Category: Brief Answer (6 points)

- Himalayan rivers originate in high snow-covered mountains, whereas Peninsular rivers originate in plateau regions or hill ranges.
- Himalayan rivers are perennial because they receive water from both melting glaciers and monsoon rainfall.
- Peninsular rivers are seasonal because their water flow depends entirely on monsoon rainfall.
- Himalayan rivers have long courses, causing deep vertical erosion and forming V-shaped valleys and gorges.
- Peninsular rivers have shorter, shallow courses with comparatively low erosion capacity.
- Himalayan rivers form large meanders and expansive deltas, whereas west-flowing Peninsular rivers form estuaries.

---

(2) Explain : Drainage Pattern and Water Divide

Answer Category: Brief Answer (6 points)

- A drainage pattern refers to the geometric arrangement formed by a main river and its tributaries across a terrain.
- Drainage patterns develop based on slope, underlying rock structure, and climate (e.g., dendritic, trellis, rectangular, radial).
- A dendritic pattern resembles tree branches, while a radial pattern forms when streams flow outward from a central mountain peak.
- A water divide is an elevated landmass, mountain range, or upland that separates two adjacent river basins.
- It prevents the water streams of two neighboring drainage systems from directly mixing with each other.
- The Aravalli Range serves as a major water divide in India, separating the Indus basin from the Ganga basin.

---

(3) Explain the utility of lakes

Answer Category: Brief Answer (6 points)

- Lakes act as natural water storage reservoirs that regulate the volume of water flowing through surrounding rivers.
- They store excess water during heavy monsoon rains to prevent severe downstream flooding in surrounding areas.
- They release stored water during dry summer seasons to maintain a continuous river flow and prevent droughts.
- Lakes are extensively used for generating hydroelectricity and supplying freshwater for domestic and agricultural needs.
- They support vibrant aquatic life, foster fishing industries, and improve the local ecological environment.
- Lakes serve as major tourist destinations, providing recreational activities like boating and water sports.

---

(4) State the remedies to prevent water pollution.

Answer Category: Brief Answer (6 points)

- Strict environmental laws must be implemented to prohibit industrial units from dumping untreated chemical waste into rivers.
- Industrial units must mandatory process chemical effluents in treatment plants before releasing water into natural bodies.
- Government initiatives under the National River Conservation Plan (NRCP) should be enforced vigorously across urban areas.
- Citizens must refrain from throwing domestic sewage, household garbage, and religious waste into rivers and lakes.
- Excessive use of chemical fertilizers and toxic pesticides in farming should be reduced to prevent agricultural runoff.
- Community awareness programs must be conducted regularly to emphasize the importance of keeping natural water clean.

---

(5) 'Godavari is named as Ganga of South' - Give reasons.

Answer Category: Short Answer (4 points)

- The Godavari is the longest and largest peninsular river system in the entire southern region of India.
- Its extensive drainage basin covers a vast catchment area spread across multiple states in peninsular India.
- It holds immense cultural, historical, and religious significance for people living in southern India.
- Because of its exceptional length, vast basin area, and spiritual value, it is called 'Dakshin Ganga' (Ganga of the South).

---

(6) Explain Indus River System

Answer Category: Long Answer (8 points)

- The Indus River originates in Tibet near Lake Mansarovar and enters India in the high-altitude Ladakh region.
- It forms spectacular deep gorges while flowing through the mountainous terrain of northern India.
- Major tributaries joining the Indus in the Ladakh region include the Zaskar, Shyok, Nubra, and Hunza rivers.
- Its five main tributaries—Jhelum, Chenab, Ravi, Beas, and Satluj—join together at Mithankot in Pakistan.
- The river flows south-west through Pakistan before finally draining into the Arabian Sea east of Karachi.
- The total length of the Indus River is nearly 2,900 kilometers, making it one of the longest rivers in the world.
- In India, its drainage basin covers parts of Ladakh, Jammu & Kashmir, Himachal Pradesh, and Punjab.
- Under the Indus Water Treaty of 1960, India is allocated 20% of the total water carried by the Indus river system.

---

2. Answer the following

(1) Explain the Ganga River System.

Answer Category: Long Answer (8 points)

- The Ganga originates as the Bhagirathi from the Gangotri glacier and joins the Alaknanda at Devprayag to form the Ganga.
- The river leaves the mountains and emerges onto the flat northern plains at Haridwar.
- Its major right-bank tributary is the Yamuna, which originates at Yamunotri and meets the Ganga at Prayagraj (Triveni Sangam).
- Key left-bank tributaries originating in the Nepal Himalayas include the Ghaghara, Gandak, and Kosi rivers.
- Peninsular tributaries like the Chambal, Betwa, and Son join the Ganga system from the semi-arid southern plateau.
- In West Bengal, the river splits into the Hooghly branch and the main river stream that enters Bangladesh as the Padma.
- The Padma merges with the Brahmaputra (Jamuna) in Bangladesh to form the mighty Meghna River before reaching the Bay of Bengal.
- Together with the Brahmaputra, it forms the Sundarbans Delta, the largest and fastest-growing river delta in the world.

---

(2) Discuss the Narmada basin.

Answer Category: Brief Answer (6 points)

- The Narmada River originates from the Amarkantak plateau in the Anuppur district of Madhya Pradesh.
- It flows westward over a length of 1,312 kilometers toward the Arabian Sea through a deep rift valley.
- The rift valley was created by geological faulting between the Vindhyan range in the north and Satpura range in the south.
- Near Jabalpur, it flows through a stunning marble rocks gorge and forms the magnificent Dhuandhar Waterfalls.
- Most tributaries of the Narmada are short and join the main river at right angles in a trellis pattern.
- Its drainage basin covers extensive parts of Madhya Pradesh and Gujarat before emptying into the Gulf of Khambhat.

---

(3) Describe in detail the Krishna and Kaveri basins.

Answer Category: Long Answer (8 points)

- The Krishna River originates from a freshwater spring near Mahabaleshwar in the Western Ghats of Maharashtra.
- It flows eastward across peninsular India for approximately 1,400 kilometers before emptying into the Bay of Bengal.
- Its major tributaries include the Tungabhadra, Koyna, Bhima, Ghataprabha, and Malaprabha rivers.
- The Krishna basin is shared among Maharashtra, Karnataka, Telangana, and Andhra Pradesh.
- The Kaveri River originates in the Brahmagiri mountain range of the Western Ghats in Karnataka.
- It flows eastward for a total length of about 800 kilometers to reach the Bay of Bengal south of Cuddalore.
- Main tributaries of the Kaveri include the Amravati, Bhavani, Hemavati, and Kabini rivers.
- The Kaveri basin covers regions of Karnataka, Kerala, and Tamil Nadu, forming the famous Sivasamudram Waterfalls.

---

3. Find out the correct option from the given options

- **(1) Which lakes are formed due to meandering of rivers ?**
  - **Answer:** (B) Ox-bow
- **(2) What is it called when a mountain or an upland separates the water flow of rivers from one another ?**
  - **Answer:** (B) Water divide
- **(3) Which of the following is not a peninsular river ?**
  - **Answer:** (C) Kosi
- **(4) Which of the following lakes is used for producing salt ?**
  - **Answer:** (B) Sambhar
- **(5) Which are the main tributaries of Ganga ?**
  - **Answer:** (A) Yamuna, Ghaghra, Gandak and Kosi